MATSAPHOUSE

3 science narration scripts

Blind spot · Recorded voice · Electric eel

Three short, child-friendly narration scripts of about 70–80 seconds, checked against public sources on 11 November 2026 (Japan time). They are ready for recording, with production cues and short source notes.

Two important corrections were made before packaging. A 1966 study summary in the recorded-voice script could not be checked in the full paper and did not match the official abstract, so that precise claim was removed. The electric-eel script no longer says the fish “out-zaps” a wall socket: 860 volts is several times a typical outlet voltage, but danger depends on current and how long it lasts, not voltage alone.

These are narration drafts, not academic papers. The notes below each script list the sources used. Check the cited work before posting if a claim is essential.

Script 01 · Neuroscience · about 75 seconds

There's a Hole in Your Vision Right Now (and Your Brain Hides It)

Each of your eyes has a blind spot where there are zero light sensors, and your brain quietly paints over it. Try the test at 0:40. #neuroscience #brain #science #matsapphouse

Hook: “Right now, there's a hole in what you're seeing, and your brain is covering it up.”

Narration and visual cues

Time	Narration	Visual cue
0:00	Right now, there's a hole in what you're seeing, and your brain is covering it up.	Extreme close-up of an eye; a black circle "punches" through the frame, then smoothly heals over.
0:06	At the back of each eye is the retina, a thin sheet packed with light-sensing cells called rods and cones.	Animated cutaway of the eyeball; retina glows pink-green, rods and cones twinkle.
0:14	But all the nerve fibres from those cells have to leave the eye somewhere. They bundle together into the optic nerve and exit through one spot: the optic disc.	Fibres converge like streams into a river and pour out the back of the eye.
0:24	At that spot there are no rods and no cones at all. Zero. So any light that lands there is simply not detected. That's your blind spot, and you have one in each eye.	The optic disc highlighted; light rays hit it and vanish with a soft "blip".
0:36	Want to find yours? Close your left eye. Stare at the cross with your right eye, and slowly move closer to the screen.	Full-screen test card: a black + on the left, a big pink dot on the right, white background. On-screen text: "Close LEFT eye. Look at the +. Move closer slowly." Hold 8s.
0:46	At one distance, the dot just... vanishes.	Hold the card; nothing changes on screen (the effect happens in the viewer's eye).
0:50	Here's the wild part: you don't see a black hole there. Your brain fills the gap using the surrounding colours and patterns.	Swap card: dot sitting on green stripes. Text: "Try again: the stripes look unbroken."
1:00	Normally you never notice, because your two eyes cover each other's blind spots, and your brain stitches it all together.	Two overlapping vision cones; each blind spot covered by the other eye's view.
1:08	So what you "see" isn't a live camera feed. It's your brain's best guess.	Presenter taps their temple, grinning. End card: matsapphouse logo.

Production notes

• The test card must be simple and high-contrast; put the + and the dot roughly 8–10 cm apart on a phone-sized frame (viewers will adjust distance).

• Say "close your LEFT eye, look at the cross" exactly; if the dot is on the right, the right eye is the one being tested.

• Keep the test card on screen long enough to try (at least 8 seconds).

Source notes

• The optic disc, where the optic nerve leaves the eye, has no photoreceptors, so each eye has a blind spot. This is standard textbook anatomy, first described by Edme Mariotte in the 1660s.

• The brain fills in the blind spot with surrounding colour and pattern (perceptual filling-in): V. S. Ramachandran, "Blind Spots", Scientific American 266(5):86–94 (1992). https://doi.org/10.1038/scientificamerican0592-86

• With both eyes open, each eye's blind spot falls on a region the other eye can see, so it usually goes unnoticed.

Script 02 · Physics of sound · about 75 seconds

Why Your Voice Sounds So Weird on Recordings

That odd feeling when you hear your own voice played back? It's physics, not a broken microphone. Part of what you normally hear travels through your skull. #science #sound #physics #matsapphouse

Hook: “Ever heard yourself on a recording and thought, 'That is not what I sound like'? You're right, and also wrong.”

Narration and visual cues

Time	Narration	Visual cue
0:00	Ever heard yourself on a recording and thought, "That is not what I sound like"? You're right... and also wrong.	Presenter plays a voice memo on a phone and looks surprised.
0:07	When other people hear you, your voice travels one way: through the air, into their ears.	Pink sound waves ripple from the mouth across the room into a listener's ear.
0:14	But when you hear yourself, the sound takes two routes. Some goes through the air, around to your ears.	The same pink waves curl around to the speaker's own ear.
0:21	And some travels straight through your body: from your vocal cords, through the bones of your skull, right to your inner ear. That's called bone conduction.	X-ray style animation: green waves travel through the skull bone directly to the spiral cochlea.
0:32	Your skull and tissues pass on the low, deep sounds especially well, so the voice inside your head sounds richer and lower than the one everyone else hears.	Graphic equaliser: the "inside your head" bars are boosted on the bass side.
0:43	A microphone only catches the air version — not the bone-conduction path. That's why a recording can sound thinner or higher than the voice you expect.	Microphone icon catches only the pink waves; green waves fade out.
0:54	Here's the surprising part: that air-only version is much closer to how you already sound to everyone else.	Split screen: "you, from the inside" versus "everyone else / the recording".
1:03	So the voice that startles you on a recording is often just one you've rarely heard from the outside — the one your friends already know.	Presenter plays the memo again, smiles, shrugs. End card: matsapphouse logo.

Production notes

• Easy home demo for B-roll: plug your ears with your fingers and hum. It sounds louder and deeper (the occlusion effect, from bone-conducted sound being trapped). Optional; keep it as a visual only if time runs short.

• Do not say that "studies prove everyone hates their voice", or that people usually fail to recognise themselves and rate the recording more harshly. Those precise claims were not kept (see source notes).

Source notes

• When you speak, you hear your voice both by air conduction and by bone conduction through the skull to the cochlea: Georg von Békésy, "The Structure of the Middle Ear and the Hearing of One's Own Voice by Bone Conduction", Journal of the Acoustical Society of America 21:217–232 (1949). https://doi.org/10.1121/1.1906501

• Bone conduction emphasises lower frequencies, so your own voice sounds fuller and lower to you than the air-only version other people hear. Recordings capture only the air-conducted sound. This is well established in audiology.

• A draft line citing Holzman and Rousey (1966), "The voice as a percept", Journal of Personality and Social Psychology 4(1):79–86, https://doi.org/10.1037/h0023518, said people "often didn't recognise their own" voice and "rated their own voice more harshly." The full paper is behind a paywall and was not available to check wording. The official abstract reports affective disturbance, a mismatch with how people expected to sound, and defensive negation — not a simple recognition-failure or harsher-rating result. A 1967 follow-up paper is the one later popular accounts cite for recognition rates, and later work is mixed (for example, people may rate an unrecognised recording of themselves as more attractive: Hughes and Harrison, 2013). Because the precise study summary could not be verified, it was removed from the narration. The script now stays with the established physics.

Script 03 · Animal kingdom · about 80 seconds

The Electric Eel Isn't an Eel (It's a Living Battery)

A South American knifefish can make hundreds of volts with stacked living "batteries." A higher voltage than a wall socket does not automatically mean a more dangerous shock. #animals #nature #science #matsapphouse

Hook: “This fish has been measured at 860 volts, and it isn't even an eel.”

Narration and visual cues

Time	Narration	Visual cue
0:00	This fish has been measured at 860 volts, and it isn't even an eel.	Dark river; a long olive-grey fish glides past; a crackle of light outlines its body.
0:06	It lives in the rivers of South America. Despite the name, it's a knifefish, more closely related to catfish than to true eels.	Map of the Amazon basin; family-tree card: "knifefish, related to catfish — not true eels."
0:15	Most of its body is packed with electric organs: thousands of cells called electrocytes, stacked like batteries in a torch so their tiny voltages add up.	Cutaway: pink-green rows of disc-shaped cells light up; a voltage counter ticks upward.
0:25	In 2019, scientists described Electrophorus voltai and measured 860 volts — the strongest shock known from any animal, and several times the voltage of a typical wall socket.	Card: "Electrophorus voltai · 860 V · de Santana et al., 2019".
0:36	But voltage isn't the same as danger. A wall socket can keep pushing a strong current for a long time. An eel's zap is a very short pulse. It can still hurt and make muscles seize up, so treat both with respect.	Simple graphic: "voltage" versus "current and time"; a brief pulse beside a continuous outlet current.
0:49	Quick shocks can make hidden prey twitch, giving away exactly where it's hiding.	A small fish hidden under leaves twitches; the eel turns and strikes.
0:57	And when threatened, an eel can leap from the water and press its chin against the attacker, sending the shock straight in. Alexander von Humboldt described this over 200 years ago; a scientist filmed it in 2016.	Illustrated recreation of a leaping eel (not a historic engraving unless public domain), then a modern lab-style shot.
1:11	A fish, a battery, and a high jumper. Nature got there first.	The eel glides away into the dark. End card: matsapphouse logo.

Production notes

• Do not use the old Humboldt engraving unless it is confirmed public domain; an illustrated recreation is safer.

• If generating footage, keep eel anatomy plausible: no pelvic fins, and a long anal fin along the belly that it uses to swim.

• Do not say the eel "out-zaps" or is "more dangerous than" a wall socket. Voltage can be higher; danger also depends on current and how long it lasts.

• Do not encourage touching eels or anything in a wall socket.

Source notes

• Electric eels (Electrophorus) are South American knifefishes (Gymnotiformes), more closely related to catfishes than to true eels (Anguilliformes). Standard taxonomy.

• Electrocytes in the electric organs are stacked in series so their voltages add up. Textbook physiology.

• Three living species are recognised. One specimen of Electrophorus voltai produced a high-voltage discharge of 860 V, the strongest known from a living animal: de Santana, C. D. et al., Nature Communications 10:4000 (2019). https://doi.org/10.1038/s41467-019-11690-z (open access). The 860 V figure is a measured discharge from one fish (1219 mm long), not an average for every eel.

• A typical household socket is about 100–240 volts depending on the country, so 860 V is several times higher. That is not the same as being more dangerous. Shock risk depends on current and how long it flows. The 2019 paper reports voltage, not a simple "more dangerous than a wall socket" result. Popular interviews with the authors describe the eel's pulse as brief, and household supply as able to deliver a stronger, longer-lasting current. The narration stays qualitative on that point and does not use unverified ampere figures.

• High-voltage pulses can make hidden prey twitch, revealing their location: Catania, K. C., "The shocking predatory strike of the electric eel", Science 346:1231–1234 (2014). https://doi.org/10.1126/science.1260807

• Leaping attack on threats, supporting Humboldt's 1800 account: Catania, K. C., Proceedings of the National Academy of Sciences 113:6979–6984 (2016). https://doi.org/10.1073/pnas.1604009113